# Triangle Inequality Theorem; Converse of the Pythagorean Theorem

*Question Bank 6.01*

Name_____
Date_____
Period_____

*Show your work. Circle your final answer.*

**For questions 1 through 3, can the three lengths make a triangle?**

**1.** 3 m, 9 m, 11 m

**2.** 5 in, 13 in, 7 in

**3.** 8 ft, 4 ft, 4 ft

**For questions 4 and 5, two sides are given. What lengths can the third side be?**

**4.** 3 cm and 8 cm

**5.** 1 cm and 5 cm; the third side is a whole number of centimeters

**For questions 6 through 8, do the lengths make a right triangle?**

**6.** 4.5 cm, 6 cm, 7.5 cm

**7.** 26 in, 24 in, 10 in

**8.** 7 m, 10 m, 12 m

**Choose one.**

**9.** Which set of lengths makes a right triangle?

- ○ A. 4, 5, 6
- ○ B. 2, 3, 5
- ○ C. 6, 8, 12
- ○ D. 11, 60, 61